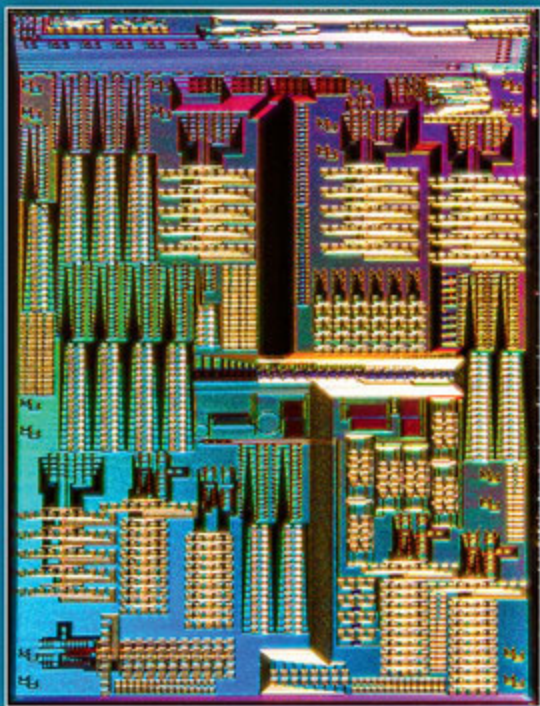


LUKAS CHROSTOWSKI
MICHAEL HOCHBERG



SILICON PHOTONICS DESIGN

FROM DEVICES TO SYSTEMS

Silicon Photonics Design

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Lukas Chrostowski is Associate Professor of Electrical and Computer Engineering at the University of British Columbia. He is the Program Director of the NSERC CREATE Silicon Electronic-Photonic Integrated Circuits (Si-EPIC) training program, has been teaching silicon photonics courses and workshops since 2008, and has been awarded the Killiam Teaching Prize (2014).

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“Photonics technology has created some of the most stunning achievements in human history, but the challenges of implementing even simple systems have made it the domain of a few specialized laboratories.

Silicon photonics enables the design of photonic systems in a much more streamlined manner, and the resulting designs can be fabricated by highly evolved silicon manufacturing facilities.

This book provides a complete guide, from physical principles of device operation through fabrication and testing, using real system examples. It gives non-specialists access to what may be the most important next step in information technology.”

Carver Mead, California Institute of Technology

“The book covers everything one would need to design, lay out, simulate, and fabricate an actual silicon chip for processing, detecting, and modulating light signals. The book’s focus on the practical side of chip implementation means that it is quite different, and frankly more useful, for chip designers than other photonics books. I highly recommend *Silicon Photonics Design* for both experienced designers and those wishing to get up-to-speed quickly in the nascent field of silicon photonics chip design.”

R. Jacob Baker, University of Nevada

“*Silicon Photonics Design* is an essential text for anyone with an interest in the application of silicon-based optical circuits, either in a commercial or academic research environment. The authors have captured all of the essential elements of silicon photonics while ensuring the text remains accessible. The inclusion of so many worked examples mixed with detailed fundamental physical descriptions is an approach that must be applauded.”

A. P. Knights, McMaster University

Silicon Photonics Design

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Preface

The academic literature on silicon photonics is sufficiently rich that one might legitimately ask whether another book in this field is needed. Certainly all of the basic physics of waveguides, modulators, lasers, and photodetectors is covered in great detail in a series of landmark texts, from Yariv and Yeh [1] to Sze and Ng [2] to Siegman [3] and Snyder and Love [4]. More specifically integrated photonics theory is covered comprehensively in texts by Hunsberger [5], Coldren *et al.* [6], Kaminow *et al.* [7], etc. Several excellent volumes have come out in recent years describing the state of the field in silicon photonics, and discussing design considerations for a variety of devices [8–15].

So what are we aiming to add to this body of literature? Our aim is not to replicate any of the existing texts' approach, but instead to provide a practical, examples-driven introduction to the practice of designing practical devices and systems. Our (admittedly ambitious) goal for this text is to do something similar to what Mead and Conway did with their landmark text on VLSI [16]: to treat the minimal possible level of device physics, and to focus primarily on the practical design considerations associated with using state-of-the-art silicon photonic foundry processes to build real, useful systems-on-chip.

In order to do this, we focus on a series of tutorials, using the tools that are in use in our own labs. That doesn't mean that these tools are perfect, or that they are necessarily the best tools for any given application: they are just what we have used. Wherever there are alternative approaches, we highlight them and provide some context for why we choose to do things in a certain way. This is obviously an area where errors of omission are very easy to make: we welcome feedback and input.

The vendors of the commercial software we use provide in-kind access for educational institutions. For example, Lumerical Solutions software is available via the Commitment to University Education (CUE) program [17], which provides access to students in undergraduate and graduate classes. Similarly, Mentor Graphics has a higher education program [18] that provides software for classroom instruction and university research. The software has been available at the silicon photonics instructional workshops we have offered.

We also provide a cursory literature review in each chapter, as well as some exercises.

Silicon photonics – training programs

This book was developed in the context of training programs in silicon photonics led by the authors, specifically in the [NSERC Si-EPIC Program](#) (Canada) [19] and the [OpSIS Workshops](#) (United States) [20].

There are and have been several training opportunities in silicon photonics around the world, including the following.

- CMC Microsystems – University of British Columbia Silicon Nanophotonics Course (Canada) [21, 22], 2007–
- [OpSIS Workshops](#) (United States), 2011–2014
OpSIS offered five-day intensive training workshops that have trained over 100 researchers and students in the design of silicon photonic systems.
- ePIXfab Europractice (Europe) [23]
- JSPS International Schooling on Si Photonics (Japan) [24], 2011
- Silicon Photonics Summer School (St. Andrews, UK) [25], 2011
- Summer School on Silicon Photonics (Peking University, China) [26], 2011–
- NSERC Silicon Electronic Photonics Integrated Circuits (Si-EPIC) Program (Canada) [19], 2012–
This program offers four annual workshop/courses each of which includes a design-fabrication-test cycle. The workshops are on the topics of: (1) Passive silicon photonics, (2) Active silicon photonics, (3) CMOS electronics for photonics and (4) Systems Integration and Packaging
- plat4M Summer School Silicon Photonics (Ghent University, Ghent, Belgium) [27], 2014

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I'd like to thank Lukas for all the work that he's put into this volume over the past few years. Lukas is a truly gifted educator, and I'm constantly impressed by his ability to communicate complex ideas to students, both in a classroom and in writing. The overwhelming majority of the work that went into this book was his, with help from a number of the students in both his and my groups.

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Chrostowski's acknowledgements

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Contributions

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Yun Wang and Li He – fibre grating couplers (Section 5.2); Dylan McGuire – model development for the modulator and detector (Section 6.2.4, 7.5); Amy Liu – scripts for the modulator (Section 6.2.4); Arghavan Arjmand – scripts for the detector (Section 7.5); Jonas Flueckiger – photonic circuit circuit simulations (Chapter 9) and automated testing (Section 12.2); Miguel Guillén Ángel Torres – directional coupler FDTD S-parameters (Section 9.4); Robert Boeck – fabrication corner analysis (Section 11.1.2) and directional couplers (Section 4.1); Dan Deptuck and Odile Liboiron-Ladouceur – design for test and check-list (Section 12.3); Yang Liu and Ran Ding – WDM transmitter (Section 13.1); Matt Streshinsky – Mach-Zehnder modulators; Kyle Murray – parasitic coupling (Section 4.1.7); Chris Cone – process design kit; Samantha Grist – SEM images; Han Yun – test setup and diagrams; and Nicolas Jaeger and Dan Deptuck – co-development of the CMC-UBC Silicon Nanophotonics Fabrication workshops and Si-EPIC workshops which formed the basis for this text.

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Part I

Introduction

1 Fabless silicon photonics

1.1 Introduction

We are on the cusp of revolutionary changes in communication and microsystems technology through the marriage of photonics and electronics on a single platform. By marrying large-scale photonic integration with large-scale electronic integration, wholly new types of systems-on-chip will emerge over the next few years.

Electronic-photonic circuits will play a ubiquitous role globally, impacting such areas as high-speed communications for mobile devices (smartphones, tablets), optical communications within computers and within data centres, sensor systems, and medical applications. In particular, we can expect the earliest impacts to emerge in telecommunications, data centers and high-performance computing, with the technology eventually migrating into higher-volume, shorter-reach consumer applications.

In the emerging field of electronics in the 1970s, Lynn Conway at Xerox PARC and Professor Carver Mead at Caltech developed an electronics design methodology, wrote a textbook, taught students how to design electronic integrated circuits, and had their designs fabricated by Intel and HP as multi-project wafers, where several different designs were shared in a single manufacturing run [1]. These efforts led to the foundation of an organization named MOSIS in 1981 that introduced cost sharing of fabrication runs with public access. The inexpensive design-build-test cycle enabled by MOSIS trained, and continues to train, thousands of designers who are responsible for the ubiquity of electronics we see today. MOSIS got started based on commercial processes that were already in production, and opened them up to the design community for prototyping and research purposes.

One of the keys to the long-term success of the microelectronics community, and in particular of the CMOS community, has been this type of access. By making these volume production processes publicly available for research and development at modest cost, anyone with a very modest level of funding is able to do cutting edge, creative work *in a process that can instantly go into large-scale production*. Training student engineers to use the production tools and processes, and then letting them loose to build cutting-edge circuits which can, with modest funding, be translated into fabless IC start-ups, has been the source of countless successful companies. It is hard to over-emphasize the difference between this and the situation in photonics (and most engineering fields), where getting from research into production involves huge barriers.

Silicon photonics is currently at the same early stage of expansion as electronics was in the 1970s, but with a major advantage for chip fabrication: existing silicon foundries

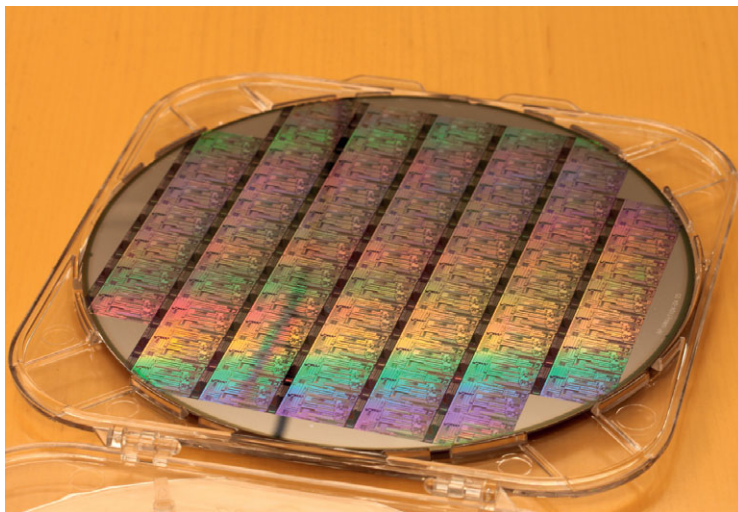


Figure 1.1 Photograph of 8" SOI wafer with various photonic components and circuits. With permission [10].

that produce highly controlled wafers (Figure 1.1) for microelectronics already exist. The microfabrication infrastructure to do silicon photonics already exists, in the microelectronics industry. Several companies are manufacturing silicon photonic chips, e.g., Luxtera's chips are already used in some high-performance computer clusters [2]. We are presently in an important transition in that academics, students, and industry worldwide now have access to active silicon photonics fabrication, e.g., via the multi-project wafer services offered by ePIXfab [3], IME [4–6], CMC Microsystems [9], and others. However, none of the processes that are currently available to the wider community are production-qualified; they are all prototyping and R&D processes, which support only very limited volumes. The inability to leverage pre-existing commercial processes was a significant impediment to the success of OpSIS, which recently shut down, since commercial users were rightly unwilling to rely on non-production-proven processes for product development. Funding from the research users and funders was not sufficient to keep the effort going, and would not allow the development of processes suitable for commercial use, which was the logical next step for the program.

We are fortunate that silicon allows us to perform all of the key optical functions at a reasonably competitive performance level, as shown in Figure 1.2, with the exception of a laser. There is a lot of recent and very interesting work going on related to growing quantum dots and germanium for lasers monolithically integrated in silicon [7]. And with the bonding technologies that have been inherited from the electronics industry, it is possible to bond lasers at relatively low cost, either through front-end integration similar to the approaches of Intel and Aurrion, or through die-scale bonding of finished laser chips [8]. These approaches are still in development, but it's clear that there are several practical approaches to making cheap light sources at various levels of integration with the silicon platform. It remains to be seen which approaches will be the most successful.

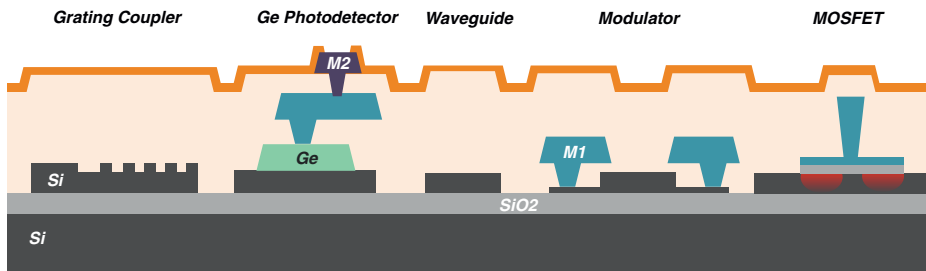


Figure 1.2 Typical process stack representing a silicon photonics platform with grating couplers, germanium photodetectors, waveguides, modulators and MOSFETs, on a silicon-on-insulator wafer. Note that most of the advanced work being done in silicon photonics today does not utilize monolithic integration with transistors, but instead leverages 3d or 2.5d integration. With permission [10].

1.2 Silicon photonics: the next fabless semiconductor industry¹

The same foundries and processes that were developed to build transistors are being repurposed to build chips that can generate, detect, modulate, and otherwise manipulate light. This is somewhat counter-intuitive, since the electronics industry spends billions of dollars to develop tools, processes and facilities that lend themselves to building the very best transistors, without any thought about how to make these processes compatible with photonics (with the exception of the processes designed to make devices such as CMOS and CCD camera chips). How are we so lucky that these capabilities can be directly reused for photonics?

In reality, they can't be directly reused. Every attempt to directly integrate photonic functionality into CMOS or bipolar silicon wafers, without making any process changes, has yielded poorly-performing devices. Electronics processes are designed for making electronics; it stands to reason they cannot be used for competitive photonics products. And even if they could, it wouldn't make economic sense; silicon photonics chips require relatively primitive processing (90 nm type capabilities) compared to advanced microelectronics chips (16 nm). Using the tools for truly advanced microelectronics to try to build photonics is a mistake, and would be impossible to justify from a performance or economic standpoint.

There is no reason to expect that the integration flows used to build electronic circuits would be in any way compatible with making components that manipulate light. But, over the last decade, it has emerged that silicon is actually a fantastic material system for building photonic devices, as well as electronic ones. And, even more surprisingly, the silicon photonics community has developed process flows that permit the re-use of CMOS fabrication infrastructure to build complex photonic circuits, where information is transferred seamlessly from the electronic to the optical domain and back again.

¹ A version of this chapter has been published in the *IEEE Journal of Solid-State Circuits* [10]: Michael Hochberg, Nicholas C. Harris, Ran Ding, Yi Zhang, Ari Novack, Choice Xuan, Tom Baehr-Jones. "Silicon photonics: The next fabless semiconductor industry", *IEEE Journal of Solid-State Circuits*, Vol. 5, No. 1, pp. 48–58, March 2013.

While the fully-integrated processes used to make transistors are not reused, modular process steps can be rearranged and reused, with distinct process flows being developed to build silicon photonics. This is not a trivial endeavor, but several organizations have shown that it is possible.

What has emerged is a vibrant community of companies and academics using the materials and techniques that have been developed over the past 50 years in the silicon microelectronics industry, and repurposing them to build photonic devices and circuits. What is particularly compelling about this work is that many of the efforts don't just make use of the same kinds of equipment in separate facilities, but actually use the exact same tools and facilities where CMOS transistors are routinely built. The constraints associated with working in such facilities are significant: materials that are not proven to be compatible with the CMOS processes are banned, and both processes and circuits have to be designed in such a way that processing them will not harm or contaminate the tools. The cost of mask sets and process development in the more advanced CMOS-compatible manufacturing facilities can also be very high [11] in advanced processes. But if the billion-dollar scale investments that go into building modern CMOS facilities can be directly leveraged to build silicon photonic systems on chip, it means that there is an immediate and a rapid path to commercialization and large-scale production is available.

1.2.1 Historical context – Photonics

Up to the present day, there has been very little in the way of opportunities for fabless photonics companies.

One of the key problems in photonics, historically, is that processes have been highly specialized for the particular application, utilizing different materials. With individual devices separately packaged and connected together by fibres, it is not unusual to see communications systems which incorporate chips made in half a dozen different material systems: RF CMOS or bipolar processes for the high-bandwidth electronics (e.g., serializers and deserializers), FPGA's or highly scaled CMOS for the digital parts (e.g., control circuits), diffused waveguides on glass for optical multiplexers (e.g., arrayed waveguide gratings) and passives, lithium niobate for modulators, indium phosphide for lasers, germanium for photodetectors, and MEMS-based switches, for instance. Each of these devices is made in a process that is fundamentally and irreconcilably incompatible with the ones used to make the other components. Each material system is chosen to provide ultimate performance for a single type of device. This means, in most cases, that the photonic components are produced in specialty fabrication facilities, in very low volumes. This results in high-cost components, since very few photonic devices are truly high-volume on the scale of the electronics industry. The only things that come close are VCSELs (which are made on a wafer scale, but are used as discrete devices), and components for PON networks (again, wafer-scale fabrication of directly modulated lasers, but they are used as singulated discrete components).

While discrete photonic devices can be interfaced to one another with standard optical fibres and connectors, a large fraction of the final device cost and yield loss

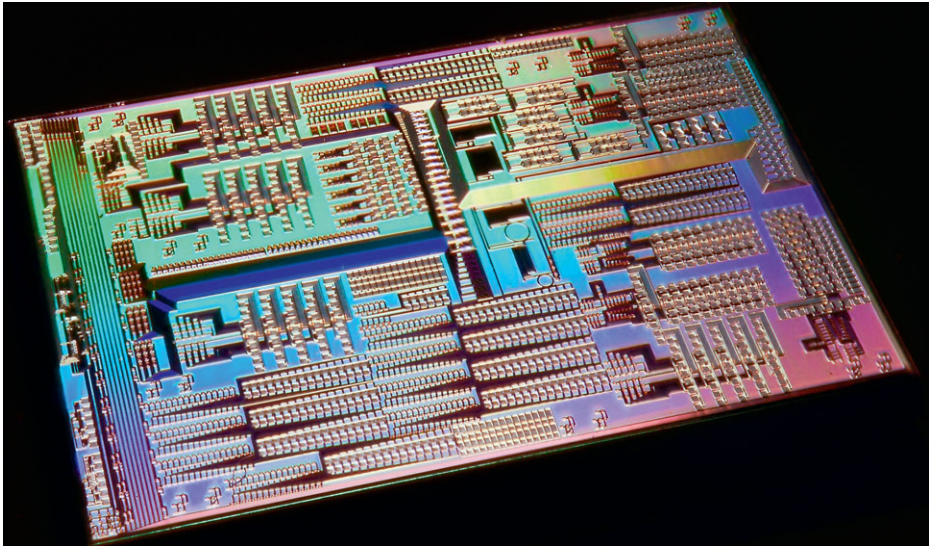


Figure 1.3 Photograph of SOI photonic chip fabricated at IME A*STAR.

emerges from the photonic packaging processes, which generally require 5- and 6-axis alignments with submicron accuracy, and from the packages themselves, which are often hermetically sealed and are sometimes quite literally gold-coated. Again, this contributes to the high cost of photonic components and systems.

The great promise of silicon photonics lies in integrating multiple functions into a single package, and manufacturing most or all of them using the same fabrication facilities that are used to build advanced microelectronics, as part of a single chip or chip stack (see [Figures 1.3-1.4](#)). Doing so will radically drive down the cost of moving data through fibres, and will create the opportunity for a variety of fundamentally new applications of photonics, where high-complexity systems can be built at very modest costs.

1.3 Applications

There are a number of applications that are emerging for complex silicon photonic systems, the most common being data communication. This includes high-bandwidth digital communications for short-reach applications, complex modulation schemes and coherent communications for long-reach applications, and so on.

Beyond data communications, there are a huge number of new applications being explored in both the commercial and academic worlds for this technology. These include: nano-optomechanics and condensed matter physics [12], biosensing [13, 14], nonlinear optics [15], LIDAR systems [16], optical gyroscopes [17, 18], radio frequency integrated optoelectronics [19, 20], integrated radio transceivers [21], coherent communications [22], novel light sources [23, 24], laser noise reduction [25], gas sensors [26], very long wavelength integrated photonics [27], high-speed and microwave signal

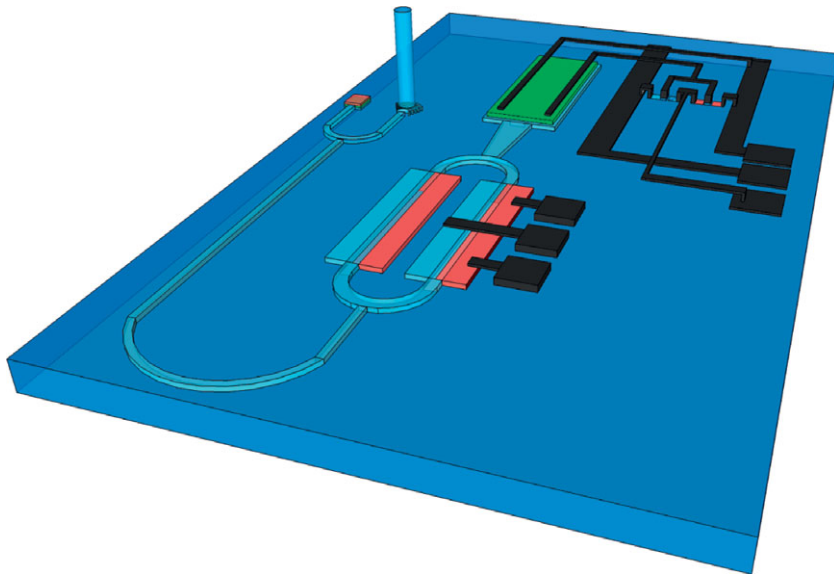


Figure 1.4 Conceptualization of a CMOS/photonic circuit. Light is coupled via on-chip laser or vertical fibre into a grating coupler. The light is then modulated, transduced by a photodetector, and inverted via a CMOS inverting circuit. Silicon photonic-electronic circuits can support systems with hundreds or thousands of such components today. Ref. [10]

processing [20], and many more. Areas of particular promise include biosensing, imaging, LIDAR, inertial sensing, hybrid photonic-RFICs, and signal processing.

1.3.1 Data communication

There are several categories of photonic devices where the silicon photonics components are competitive with best-in-class devices, discussed below. But what we regard as perhaps the most transformative work being done in the optical communications field is concerned with creating integrated platforms with modulators, detectors, waveguides and other components on the same chip, all talking to one another. In some cases, transistors are also included in these platforms, allowing amplifiers, serialization, and feedback to all be integrated onto the same chip. Because of the expense of developing such processes, this effort has largely been led by commercial players, mainly aimed at applications in point-to-point data communications. And because of the expense of developing transistor manufacturing processes, the emerging consensus in the field is that integrating electronics through bonding, either at the wafer or at the die level, makes the most sense for the foreseeable future from both a performance and cost perspective.

There is a lot of obvious value in being able to make chips that can perform computations with electronics, and transmit data optically. The vast majority of the early applications of silicon photonics are in digital data communications. This is driven by the fundamental differences in physics between electrons, which are Fermions, and photons, which are Bosons. Electrons are great for computation, because two of them

cannot be in the same place at the same time. That means that they interact very strongly with each other. As a result, it is possible to build massively nonlinear switching devices – transistors – using electronics.

Photons have a different set of properties: many photons can all be in the same place at the same time, and except under very special circumstances they don't interact with each other. That's why it is possible [28] to transmit literally terabits of data per second through a single optical fibre: this is not done by creating a single terabit-bandwidth stream of data. Instead, typical high-bandwidth fibre optic transmission systems today make use of a variety of techniques to take a large number of streams of electrical data, at 10, 28 and 40 Gbits/second speeds, and multiplex them together onto single fibres. One such technique, commonly used in telecommunications, is wavelength-division multiplexing [29], where each colour of light is modulated separately and the various colours are all combined onto a single fibre. Other techniques, where coherence is exploited in order to encode information into phase, amplitude and polarization separately, have become common in long-haul and metro telecommunications as well [30]. In fact, these different approaches can be combined, with each wavelength representing a separate stream of data, and also being modulated in phase, amplitude and polarization, in order to transmit many bits per symbol. Thus it is possible to build systems that transmit terabits of data through a single fibre, without any electronics operating faster than 28 or 40 Gbits/second.

These techniques are critical, because fibre optic data transmission over long distances is limited by the expense of laying the fibres: Obtaining right-of-way and physically laying the conduits is expensive on land, and frightfully expensive across oceans [31]. In such systems, even if it requires millions of dollars of equipment at the end-points to make efficient use of existing fibres, that is economically sensible, since the cost of laying more fibre is prohibitive. What we've seen over the past 50 years is that, because of the very low losses of fibres and the availability of erbium-doped fibre amplifiers [32] (which amplify all of the different optical wavelengths in a given window, around 1.5 μm) fibre optics first became the dominant technology for transmitting data over long distances, between cities. Over time, it came to dominate metro links, and is now dominating the relatively short distances within the data center. In many parts of the world, fiber-to-the-home is the dominant access paradigm, though this has not proven to be true in the United States, where it competes with DSL and other technologies. With the constant demand for increasing bandwidth [33], demand for ever-more-efficient ways of pushing data through fibres has grown steadily.

The broad trend in the data communications market is that as distances get shorter, the price per part drops precipitously, while the volumes go up. Unsurprisingly, silicon photonics commercialization efforts have focused a lot of effort onto the higher-volume, shorter-reach applications, aiming at data centres and high performance computing. Future applications will include board-to-board, short reach connections on the scale of USB, and perhaps eventually CPU core-to-core communications [34], though the case for on-chip core-to-core applications remains quite speculative.

Though not yet on the scale of the CMOS industry, silicon photonics is beginning to be a significant industry in its own right. The first commercial products integrating

chip-scale electronics and photonics have recently hit the market [35], and Intel has announced its intention to standardize a format for optical data communication for personal computing [REF-INTEL]. Luxtera announced the sale of their millionth silicon photonic data channel [37], and they are now selling a 100 Gbits/second-class optically active cable (4x28G) [38], fabricated using the Freescale CMOS foundry in Texas [39]. Numerous startups and established companies (Kotura, Luxtera, Oracle, Genalyte, Lightwire/Cisco, APIC, Skorprios, TeraXion, and many others) are actively developing silicon photonic products. Many of the top American defense companies have programs in this area. Moreover, many of the major semiconductor players now have active programs in silicon photonics. Intel, Samsung, IBM, ST and many others have publicly announced activity in this area. Although there have been some over-blown predictions by market research firms [40] claiming that the field will be generating \$2B/year in revenue by 2015, it does seem likely that revenues around \$1B will be achieved before 2020, as predicted by the authors [11].

1.4 Technical challenges and the state of the art

1.4.1 Waveguides and passive components

There are a wide variety of waveguide geometries that have been developed in silicon-compatible systems; nearly any transparent material with a higher refractive index than glass can be deposited on top of an oxidized silicon substrate and turned into a waveguide. For purposes of CMOS process compatibility, however, the community has converged on a few classes of geometries. The most common are high-confinement waveguides made out of the active device layer of an SOI wafer, etched either fully to the bottom oxide layer or partially etched with a timed process stop [41, 42] (Figure 1.2 and 3.4). It took several years of work to reduce the losses of these sub-micron waveguides to acceptable levels, since the strong interaction of the optical fields with the sidewalls can lead to substantial losses, driven by roughness [43]. Propagation loss can be reduced either by process optimization to smooth the sidewall [43] or by waveguide geometry optimization to reduce modal field strength at the sidewall [44]. Typical losses for high-confinement guides are in the 2 dB/cm range today for cutting-edge processes [45]. Low loss multimode straight waveguides in combination with tight single mode waveguide bends turn out to be an optimal choice for routing, achieving 0.026 dB/cm [46]. Other key passive components such as grating couplers [47] (Figures 1.2, 1.5, and 1.6), distributed Bragg gratings [48], waveguide crossings [49], and arrayed waveguide gratings (AWG) [50] have all been demonstrated, in each case with very low losses. More recently, CMOS compatible waveguides that can be formed into the dielectric back-end process have become available, made out of silicon nitride. With dedicated processing, the losses of these waveguides are extraordinarily low (< 0.1 dB/m), though the compatibility of such processes with front-end active devices is an open issue, given the requirement for high temperature growth [51]. It should be noted that considerable work has been done on low-confinement silicon waveguides [52, 53].

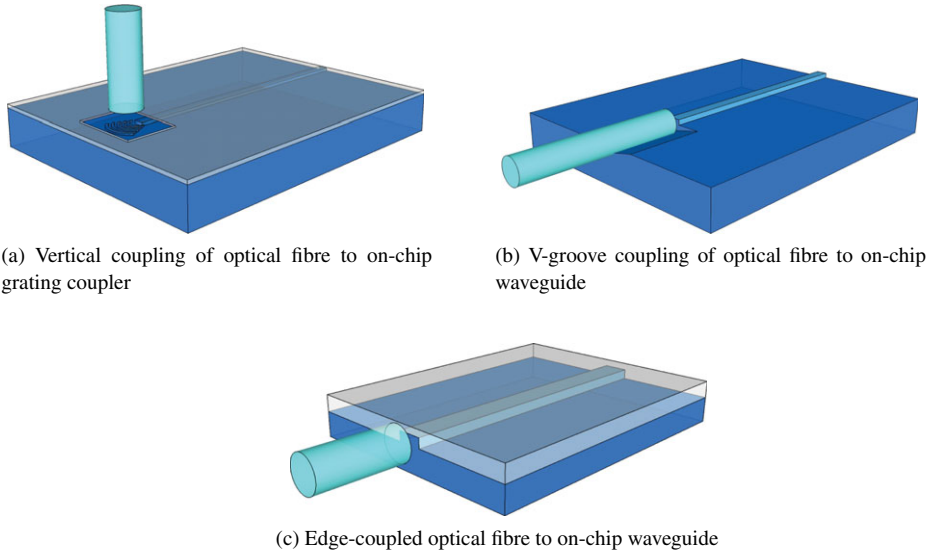


Figure 1.5 Light coupling techniques. With permission [10].

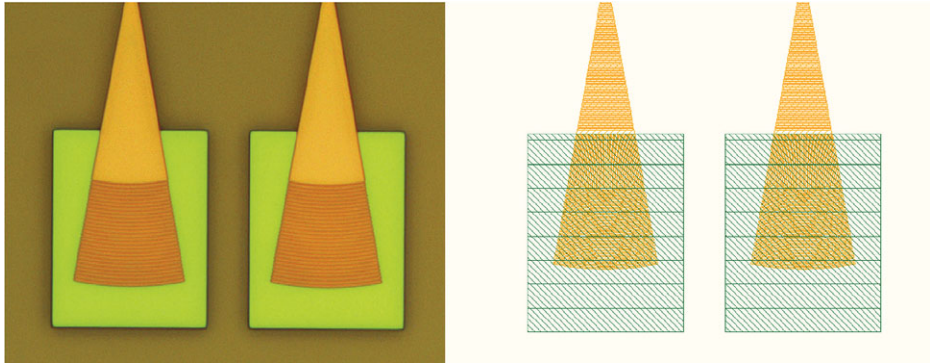


Figure 1.6 Optical micrograph and layout of an optical grating coupler fabricated at IME A*STAR. With permission [10].

There is considerable difficulty in creating compatible high-speed modulators and detectors in these platforms, making them less promising for large-scale integration.

One of the challenges with silicon photonics is coupling light between the chip and optical fibres, and achieving this is in a packaging method that is cost effective [54]. This is typically done using edge or grating couplers, as in Figure 1.5, and described in Chapter 5. Both approaches have demonstrated below 1 dB loss per interface [54–56]. Dealing with polarization is also a challenge, since silicon photonic waveguides are, by default, highly birefringent, namely the optical propagation constants in the waveguides are different for the two different polarizations. The common approach is to build circuits using a single polarization, and to duplicate them when both polarizations are

required. This is termed polarization diversity, and takes advantage of polarization-splitting grating couplers [57], polarization splitters [58], splitter-rotators [59], or other related components. Other approaches (ref kotura?) have been explored which rely on using square waveguides to eliminate birefringence, but these approaches impose significant design constraints.

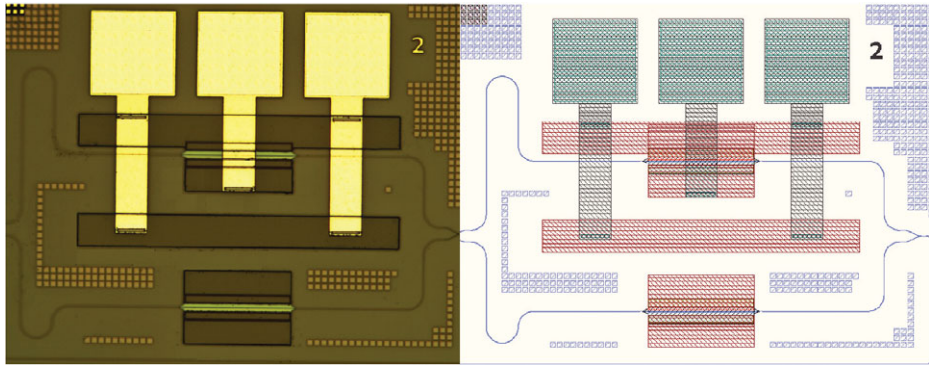
1.4.2 Modulators

Modulation in silicon is most commonly achieved by the plasma dispersion effect [60], where free carrier density changes can induce changes in the refractive index and modulate the light. Several different mechanisms of manipulating free-carrier densities have been implemented in monolithic devices [61]. Among them, carrier-depletion mode devices, usually based on a reverse-biased pn junction, are widely used to achieve high-speed operation.

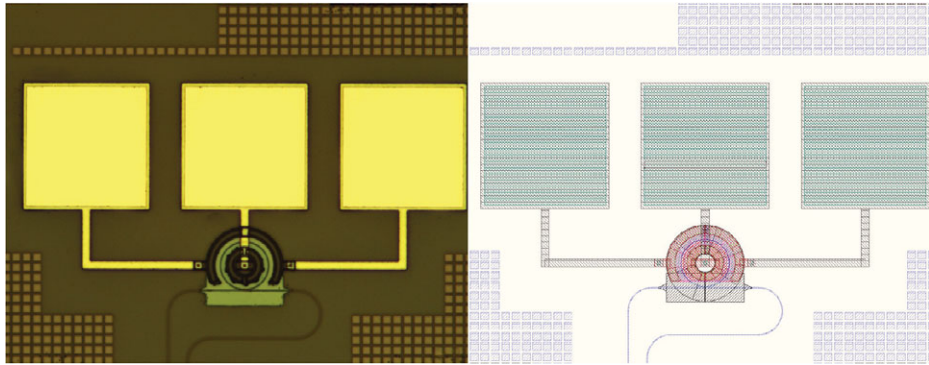
Since the first GHz silicon modulator was demonstrated by the Intel group [62], the modulators' performance metrics have been significantly improved. Mach-Zehnder interferometer (MZI) structures [63] are typically used for amplitude modulation. With traveling-wave designs, data rates up to 50 Gbit/s have been experimentally demonstrated [64,65]. At 20 Gbit/s, a record low power consumption of 200 fJ/bit has recently been reported [66]. An example MZI modulator is shown in [Figure 1.7a](#).

Resonant structures can be used to dramatically reduce the device footprint and further reduce power consumption, although at the cost of significantly narrower operating wavelength window and high thermal sensitivity. First introduced by Xu et al. [42] in 2005, high-speed ring modulators have been demonstrated to operate up to 40 Gbit/s [67] and recently were reported to achieve 7 fJ/bit at 25 Gb/s with thermal tuning capabilities [68]. An example of a ring modulator is shown in [Figure 1.7b](#), and described in [Chapter 6](#). Recent development includes breaking the cavity photon lifetime limit by coupling modulation [69], and building WDM transmitters using ring modulators, as described in [Section 13.1](#).

In addition to pure silicon solutions, other materials can be integrated on to the silicon platform, for example by bonding III/V material [70], epitaxially growing germanium [71] or cladding graphene [72] to build efficient electro-absorption modulators. Chemically engineered active electro-optic polymers have also been introduced to the silicon platform in slot waveguides [73] and photonic crystals [74] to build efficient phase shifters. The broader field of hybrid integration of new materials into silicon, either through post-processing or through various kinds of encapsulation approaches to make the new materials acceptable to a CMOS foundry, is emerging as an active subfield in silicon photonics. These approaches all tend to involve challenging fabrication, and are likely to be limited to specialized applications with very specific high-end requirements. The possible exception to this is in direct integration of light sources, though the jury is out on the various technological approaches to this problem. It remains unclear whether there is a technological or economic imperative driving the integration of the light source onto the same chip as the modulators, detectors, and other components.



(a) Mach-Zehnder modulator



(b) Tuneable ring resonator modulator

Figure 1.7 Optical micrograph of devices fabricated at IME A*STAR as well as their layouts. With permission [10].

1.4.3 Photodetectors

At the operating wavelengths of a silicon photonic chip, materials with a narrower bandgap than silicon need to be integrated to function as the detection (absorption) medium. Germanium can be grown epitaxially and germanium absorbs light at the telecommunication wavelengths [75]. This is necessary for compatibility with standard infrastructure, but not necessarily required for short-reach applications where both ends of the link can be defined, without adhering to interoperability standards. Bonded III/V materials are also used for photodetection [76]. These materials are integrated close to, or directly connected to, a silicon waveguide, so that the guided light can be evanescently coupled or butt-coupled into the photodetector and the photodetector can have a small cross-section to reduce device capacitance and improve speed [77]. The state-of-the-art for germanium photodetectors, in a photodiode configuration, is a 120 GHz bandwidth with 0.8 A/W responsivity [77]. A responsivity of 1.05 A/W was demonstrated at 20 GHz for the wavelength 1550 nm [78], which amounts to 84% quantum efficiency. An example germanium detector is shown in Figure 1.8. A very low photodetector capacitance of 2.4 fF has been achieved in a photoconductive device with

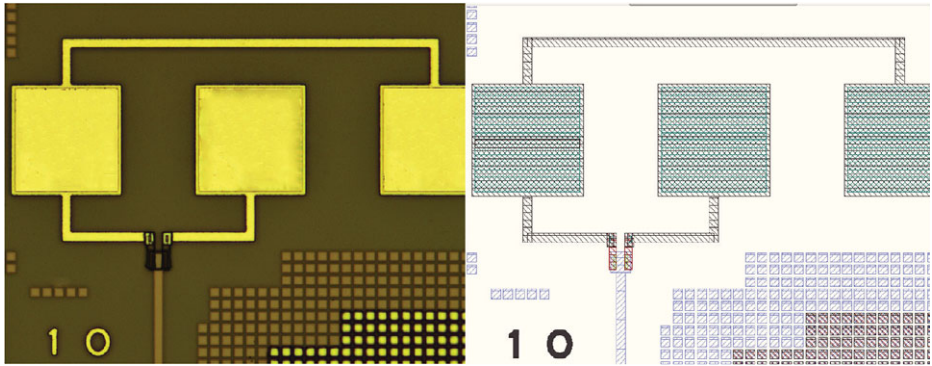


Figure 1.8 Optical micrograph and layout of a photodetector fabricated at IME A*STAR. With permission [10].

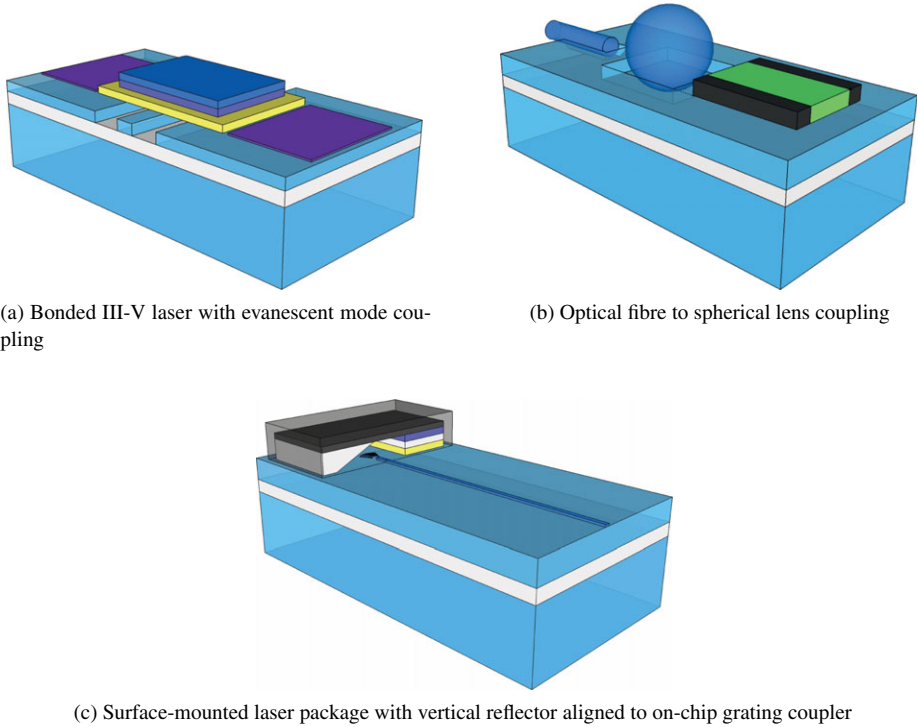
an estimated quantum efficiency of 90% operating with a bandwidth of 40 GHz [79]. Low bias avalanche detection with 10 dB gain and improved noise performance was achieved by IBM with a speed exceeding 30 GHz [80]. Detectors are discussed in detail in [Chapter 7](#).

1.4.4 Light sources

One of the main challenges of the silicon photonics platform is the lack of an on-chip light source. The current generation of silicon photonic chips in production all couple light from an external laser. While edge and grating couplers have both seen improvements in coupling efficiency, the lack of an on-chip source limits the potential applications of these chips.

A number of techniques have been proposed to address the light source issue; these are discussed briefly here, and in [Chapter 8](#). Hybrid Silicon lasers have been demonstrated using both bonding [81, 82] and epitaxial growth [24] to transfer III-V materials to the silicon wafer ([Figure 1.9a](#)). However these techniques are hindered by the incompatibility of III-V materials with the standard CMOS process along with the high cost and low yield of bonding, and the small size of the available III/V wafers. Germanium has been proposed as a CMOS compatible gain medium though its photo-emission efficiency is hindered by its indirect band gap. The small (134 meV) difference between the indirect and direct band gaps can be overcome by a combination of strain and heavy n-type doping [83] and just recently, the first electrically driven laser using germanium gain medium on silicon was demonstrated [23].

All of the products currently in the market use more pedestrian approaches (e.g., [Figure 1.9b–Figure 1.9c](#)). These include both off-chip light sources, connected to the silicon chip by fibres, and lasers integrated within the same package as the silicon photonic chip. Technologies for this kind of integration have been inherited from the MEMS community, including micro-packaging techniques, and are both low-cost and very mature. These types of co-packaging approaches are already proven in medium scale production at Luxtera, for example.



(a) Bonded III-V laser with evanescent mode coupling

(b) Optical fibre to spherical lens coupling

(c) Surface-mounted laser package with vertical reflector aligned to on-chip grating coupler

Figure 1.9 Laser integration techniques. With permission [10].

1.4.5 Approaches to photonic–electronic integration

Bringing electronics physically close to photonics can significantly improve system performance, reduce cost and open up tremendous new system design opportunities. Integration approaches generally fall into two categories: monolithic and multi-chip.

Monolithic integration

As the pioneer in monolithic electronics-photonics integration, Luxtera developed a CMOS-photonics foundry process based on a 130 nm CMOS process at Freescale and brought the first silicon photonics product to the market [35], and they recently teamed with ST Microelectronics on a development project aimed at bonded integration (see below) [84]. In their current process, the electronics and photonic components are fabricated together, in a single process flow. This required considerable process development effort, with additional layers being inserted in the flow in such a way as to allow the existing CMOS transistors to continue working, while adding germanium for photodetectors and silicon etches to support the fabrication of waveguides. Another process, from IHP, is integrating bipolar transistors with silicon photonics; this is a promising approach from a technical perspective. It remains to be seen whether cutting-edge bipolar transistors can be integrated with silicon photonic without performance compromises; if so, this could be a very powerful technology. Thus far, the integration efforts have centred around non-leading-edge bipolar transistors.

Another approach to front-end integration is to use unmodified advanced electronics process nodes, such as a 45 nm SOI CMOS process, and to attempt to integrate photonic devices within the constraints of the design rules [85]. This is an attractive approach at first blush, but the constraints imposed by the inability to modify the process mean that wavelengths are highly constrained to being near the silicon absorption edge by the low-percentage silicon germanium. Fast photodetectors and modulators have yet to be proven in these technologies, which are as yet in the early stages of development. We do not expect that this approach will yield competitive commercial products in the foreseeable future.

Multi-chip integration

On the other hand, multi-chip integration is being investigated and demonstrated, in which electronics and photonics circuitries come from different processes and are electrically bonded to each other (1.10). These can be stacked in different orders depending on the applications, e.g., the photonic chip on top with the electronics on the bottom, or vice-versa. Proposed and demonstrated approaches include using wire-bonding [86], flip-chip bump bonding (providing reduced parasitic capacitance and higher density than wire-bonding) [87], low-capacitance copper pillar interconnects, and through-silicon vias (TSV) [88] that offer even higher density and lower parasitics. The benefits of these approaches are significant: as the CMOS world continues to improve with Moore's law, it will be possible to rapidly adapt each successive generation of CMOS or Bi-CMOS onto an inexpensive silicon photonics wafer, getting the benefits of best-in-class processes for both the photonics and the electronics. Because the photonics generally does not require the fabrication of very small structures (~ 100 nm critical dimensions are typical) and devices are large compared to transistors, economic

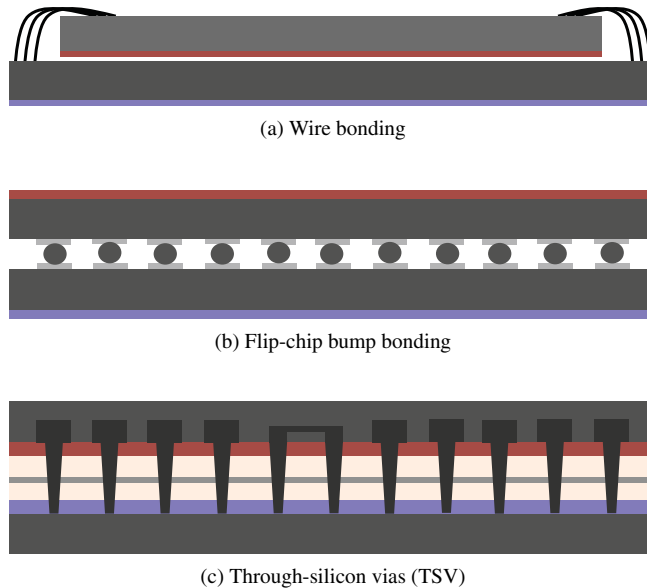


Figure 1.10 Electronic/Photonic chip-to-chip bonding schemes. With permission [10].

considerations will tend to drive the photonic devices to be made in a separate process from any advanced electronics that are needed in a final product. We expect these types of bonding processes to figure prominently in the next few years of silicon photonic research and product development.

1.5 Opportunities

We describe five fundamentally different kinds of opportunities in silicon photonics:

1.5.1 Device engineering

The first opportunity, *device engineering*, is the one that has driven perhaps 90% of the work in the field to date – particularly in terms of academic publications. Device engineering centres around answering a simple question: what are the best photonic devices that can be built in silicon?

We are now at the point where silicon Mach-Zehnder modulators are competitive with lithium niobate for many applications, and the silicon modulators continue to improve by significant factors with every passing year. By the same token, germanium waveguide-coupled photodiodes are competitive, on technical merits, with other uncooled photodetectors in the near infrared. Some of the world's lowest-loss waveguides are in silicon-nitride, which is fundamentally compatible with large scale photonic integration. Low-loss fibre couplers, various kinds of high-performance passive optics and even efficient lasers have been demonstrated within the silicon system, with co-packaging of III/V materials. More recently germanium lasers have been demonstrated, but thus far their efficiency limits their competitiveness. Much of the work on device engineering has centered around efforts at universities, because they have significant advantages in that they can close the simulation-fabrication-test loop very quickly for prototyping.

It is in the nature of device-level research to generate results that are not fully compatible with existing integrated processes. Those designing the latest and greatest new transistor may not initially be concerned about whether it fits into an existing process flow. The same is true in photonics: there will naturally be a pipeline from new devices being proven in specialty processes, to these new devices making their way into integrated platforms over time.

1.5.2 Photonic system engineering

The second major opportunity is in *photonic system engineering*. Once there is a supporting infrastructure that provides stable processes with a variety of photonic devices already included as accessible library elements, the question becomes: "What can we do with these library elements to build something useful?" Obviously one answer is to simply take single devices and package them as discrete components, but this approach does not really leverage the core advantage of silicon processing, which is to enable a rapid scaling of complexity.

In the electronics world, circuit designers do not need to be experts on the physics and fabrication of the transistor: instead, they rely on phenomenological models (termed “compact” models in the electronics industry) implemented in an environment like SPICE or VERILOG-A to simulate complex systems. Since the device models are guaranteed by the foundry, looking ‘inside’ of the device models is only required if a design is pushing the ultimate limits and needs something unusual and special. This kind of work is done in only a very small fraction of the electronic chip designs. In general, the device physicists work for the foundries, and develop the PDKs; very few foundry users will ever run the kind of TCAD simulations required to simulate the innards of the transistors.

A major near-term goal for silicon photonics foundry service providers is to make this same kind of infrastructure available for silicon photonics. Typical silicon photonic foundry service providers offer design kits with fixed designs for leading-edge photonics devices like modulators and detectors. The hope is that users will take these devices and either use them as starting points to build their own high-performance devices, or to use copies of these devices as part of complex systems containing numerous components (Figure 1.11). Photonic circuit modelling is discussed in Chapter 9.

One particularly rich area for innovation in this area is in photonics-electronic integration. Once transistors and photonics components are available in the same platform, with low-capacitance interconnects between them, it becomes possible to build cutting-edge devices that blend the functionality of what were formerly discrete devices. Luxtera, a company co-founded by one of the authors, has shown a number of results in this direction.

A transition from devices to systems

One of the most exciting aspects of the OpSIS-IME multi-project wafer shuttle runs was that many of the users, when presented with a library of functioning device designs,

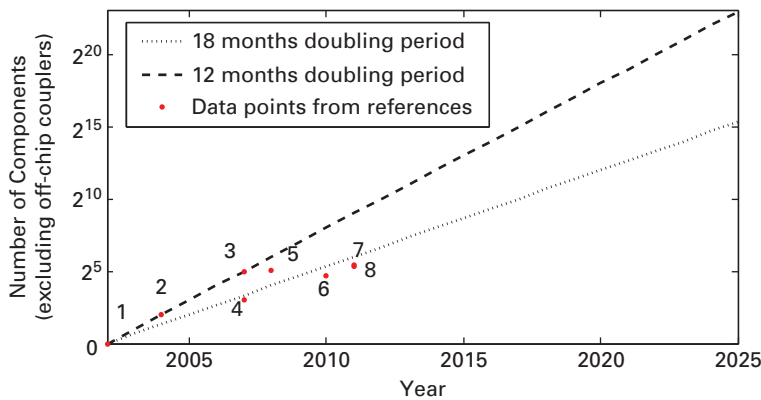


Figure 1.11 Plot of component number data points extracted from various publications on silicon photonics system design. (1) Luxtera: first silicon photonic system, (2) Intel: first CMOS modulator, (3) Kotura: VOA array, (4) Luxtera: 40Gb/s photonic chip, (5) Intel: 200 Gb/s PIC, (6) Intel: 4-channel WDM link with hybrid lasers, (7) Alcatel-Lucent Bell Lab: 10 x 25 Gb/s chip and (8) UC Davis group: fast-reconfigurable filter. With permission [10].

chose to focus their efforts on building integrated systems rather than on modifying the library elements. This was a sea-change in the field, because it meant that the basic devices were “good enough” to stop iterating on them at a device level, and shift focus toward building application-specific photonic integrated circuits (AS-PICs). This represented a very exciting moment for the field, as we moved from focusing most of the effort on device innovation into a period of very rapid growth, driven by the desire to build complex chip-scale systems.

Device innovations will continue to be important, but we can expect that over the next few years, the size of the community of people designing systems will grow much more rapidly than the community of device designers. The ecosystem now exists to build AS-PICs and to have a reasonable chance of them working in the first tape-out, at least to the point of developing prototypes. This is a new capability, and one that we expect will spark the rapid growth of a fabless photonics design community. The manufacturing transition remains riskier, and a major focus of efforts over the coming years will need to be on creating a smooth transition directly from prototype to production, by opening production processes for use as a prototyping vehicles.

1.5.3 Tools and support infrastructure

A third area of opportunity is in *developing tools and support infrastructure*. The silicon photonics design ecosystem is still very much in the development phase. Device-level simulation is now a very mature field, but higher-level simulation is just beginning to emerge. Wafer-scale automation of testing, design rule checking, layout-versus-schematic checkers, and design-for-test tools either do not exist or are extremely immature. Design-of-experiment for test structures to easily extract critical second order parameters like nonlinearity, power handling, coherence length, SFDR, etc. is an open problem in the literature.

Electronic–photonic co-design

One of the greatest innovations in the microelectronics world was the development of process design kits, PDKs, that allow design to be almost completely separated from fabrication, and which abstract the interaction between the two. There is a great deal of active work in the silicon photonics world in developing dedicated tools and design flows that can ensure first-silicon success, but these efforts are relatively immature. Luxtera was the first to develop an advanced design kit, living in a Cadence environment, which includes layout-versus-schematic and design rule checking for both the photonic and electronic components, plus statistical corner models for their standard components. Dedicated simulation tools for integrated photonics at the TCAD level have been developed by a number of companies [89–92], and existing tools for thermal and RF simulation [93] are routinely reused for simulating these aspects of photonic devices. In terms of photonic system simulation, tools include OptiSPICE by Optiwave [94], VPItransmissionMaker and Lumerical INTERCONNECT [95]. The design tools are described in more detail in [Chapter 2](#).

One of the key advantages of electronic-photonic integration is the ability to integrate thermal feedback and management into photonic chips at the device level. Doing so is

likely to dramatically simplify functional design, since thermal changes of a few degrees can radically alter the performance of photonic devices, particularly if they make use of on-chip resonant effects.

DFM and yield management

One significant area for innovation will be in developing design-for-manufacturing rules and methods, and in designing the right structures and procedures for yield management. What are the key structures to test to ensure photonic yield? What are the right ways to implement optical proximity correction in these systems? How do you map simple and inexpensive tests onto photonic system-on-chip performance? How does one pick the right mask technology for a given product, and predict the effects on performance? There is very sparse information in the public literature to address these questions, which will be critical issues over the coming years. We know empirically that even without doing any work in yield control, we routinely build wafers containing devices that operate within relatively tight tolerances (couple of dB in the electrooptic S-parameters) [6]. These numbers are quite good by optical device variation standards, but lot-to-lot data does not exist in the public domain.

1.5.4 Basic science

The fourth opportunity is in the area of *basic science*. There has been a lot of really exciting basic work in opto-mechanics and quantum optics, for instance, enabled by the high field concentrations and low losses of the silicon system. Work on novel materials like polymers, vanadium oxide, graphene, quantum dots, and many others, is ongoing, leveraging silicon photonics for various purposes. As these processes become more available over the coming years, we can expect photonics-enabled silicon to become a platform for exploring many areas of basic materials science and physics, much like VLSI electronics has proven to be an enabling technology across the sciences and engineering. A considerable amount of science in quantum optics [96] and low-temperature physics is ongoing [97]. Furthermore, silicon photonic circuits are being used as testbeds for exciting network modeling work, because of the ability to rapidly prototype new kinds of switches and transceivers [34].

1.5.5 Process standardization and a history of MPW services

A key innovation in the microelectronics industry has been the development of multi-project wafer (MPW) services, which make advanced processes available to the wider community. Making production processes available to the research and development community has been a key part of the fabless ecosystem in microelectronics. MOSIS has been an essential organization in making these processes available and accessible over the past several decades [98].

As is the case in the microelectronics world, a common commercial, production-ready foundry platform provides the following advantages:

- (1) **Resource Efficiency** – eliminates the need for in-house cleanrooms and costly process development.

- (2) Design Re-Use – the electronics industry benefits by reusing building blocks termed Intellectual Property (IP) cores rather than designing systems (such as processors) from the transistor up for each new design (over 1 billion transistors in some cases). Similarly, the same concept could allow our community to develop libraries of components. This will facilitate systems-level research and development.
- (3) Resource Accessibility – makes the fabrication process and the libraries available to the vast community of photonics and electronics designers.
- (4) Transferability for commercialization – allows industry to use the components and systems developed by the community to design new products that will be fabricated by a common foundry.
- (5) Encourages collaborative design – facilitated by common goals, language and fabrication processes.

ePIXfab and Europractice

The first publicly available silicon photonics multi-project wafer foundry services were offered by imec (formerly the Interuniversity Microelectronics Centre; in Leuven, Belgium) and CEA-Leti (Grenoble, France) via ePIXfab beginning in 2006 [3,99]. Numerous academic and industrial users worldwide have benefited from access to silicon photonics fabrication. Initially, the fabrication process was a passive silicon photonic process using 193 nm deep-UV lithography. In 2012, CEA-Leti began offering a fabrication process with heaters, to enable thermo-optic silicon photonic devices. ePIXfab then offered two further fabrication services: one offered modulators fabricated at imec, and the other offered fabrication of detectors at CEA-Leti. In 2013, ePIXfab announced a full platform (passives, modulators, detectors) with fabrication at both imec and CEA-Leti [3]. ePIXfab also offers fabrication at IHP GmbH (Innovations for High Performance Microelectronics; in Frankfurt, Germany), VTT (Finland), and packaging services at Tyndall National Institute, University College (Cork, Ireland). Silicon photonics technologies are also available via the Europractice IC Service [100].

IME

The Institute of Microelectronics (IME) is a member of the Agency for Science, Technology and Research (A*STAR), in Singapore. IME conducts research and development in silicon photonics, and has made its fabrication process available to worldwide academic and industrial customers [4]. IME has a wide range of fabrication capabilities and services which allows their customers to tailor the process based on their needs. It has been used, for example, by OpSIS and CMC Microsystems, for MPW runs.

OpSIS

Optoelectronic Systems In Silicon (OpSIS) [6], which was led by one of the co-authors and by Thomas Baehr-Jones, was a non-profit Silicon Photonics foundry access and state-of-the-art PDK design service operated by the University of Delaware in the United States. It offered the first public MPW run to include passives, modulators, and detectors on a single platform, with the fabrication performed by IME in Singapore, in 2012.

OpSIS provided high-performance silicon photonics cells and design tools, including an advanced Process Design Kit (PDK) with some of the world's highest performance active, passive and thermal photonic devices. OpSIS also provided an eco-system of service and equipment providers (design, simulation, fabrication, and packaging) for the users, simplifying the development and testing of silicon photonic device and system prototypes. OpSIS' organizational mission was to prove that photonic MPW runs could support a large community of users; at peak OpSIS had a couple of hundred active users. It offered six MPW runs, ending in 2014, when OpSIS' 5-year core funding expired and the organization shut down.

CMC Microsystems

CMC Microsystems is Canadian government funded centre for the creation and application of micro- and nano-system knowledge by providing a national infrastructure for excellence in research and a path to commercialization [101]. CMC Microsystems supports access to international foundries for the manufacturing of silicon photonics and CMOS electronics, access to packaging technologies, and access to design software. CMC supports the Canadian industry and academic institutions in their design cycles including by offering training such as workshops offered by the Si-EPIC program.

Other organizations

There are several other organizations around the world involved in silicon photonics fabrication, including offering multi-project wafers and/or providing access to fabrication. Numerous universities, organizations, companies, and institutes, are involved in silicon photonics research and have contributed immensely to the growing field of silicon photonics. While the above list is not exhaustive, it identifies that there is an international critical mass of foundries, service providers, software vendors, designers, companies, universities, etc. This growing list provides both confidence and several sources for manufacturing services to the designer seeking to develop silicon photonic systems.

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